

Feature Affinity based Clustering for Test-Time Adaptation for Image Quality Assessment

Meghna Kapoor

Dept. of Electrical Engineering
IIT Jammu, India
meghna@iitjammu.ac.in

Vinit Jakhetiya

Dept. of Computer Science and Engineering
IIT Jammu, India
vinit.jakhetiya@iitjammu.ac.in

Badri Narayan Subudhi

Dept. of Electrical Engineering
IIT Jammu, India
subudhi.badri@iitjammu.ac.in

Ankur Bansal

Dept. of Electrical Engineering
IIT Jammu, India
ankur.bansal@iitjammu.ac.in

Weisi Lin

College of Computing and Data Science
Nanyang Technological University, Singapore
wslin@ntu.edu.sg

Abstract—Recently, Test-Time Adaptation (TTA) algorithms have gained traction in image/video quality assessment (IQA/VQA). These methods use virtual losses, like group contrastive and rank loss, as auxiliary tasks to adapt batch normalization parameters, helping models generalize better to distribution shifts between training and testing datasets. Group contrastive loss clusters images into low- and high-quality groups based on predicted quality scores, maximizing feature space distance between them. However, its effectiveness relies on the base model's ability to accurately predict quality scores, which can be compromised when distribution shifts occur, leading to suboptimal adaptation and degraded performance. We propose a novel clustering approach based on the assumption that high-quality images contain richer high-level information, which is extracted using a pre-trained VGG-16 model. Images are clustered by comparing the VGG-16 features of the highest quality image in a batch with the others, enabling effective grouping based on feature affinities. These accurate clusters enhance the computation of contrastive loss, improving the adaptation of batch normalization layers. Additionally, we introduce an adaptive rank loss to reduce the impact of rank loss when the base model can distinguish images with varying distortion levels. Experimental results across multiple image quality assessment datasets, including LIVE, CID-2013, KONIQ-10K, and SPAQ, as well as algorithms like MetaIQA, HyperIQA, TRaS, and MUSIQ, show that the proposed method consistently performs better than the existing Test-Time Adaptation (TTA) approach.

Index Terms—Test-time adaptation, Image quality assessment, Batch normalization layer, Distribution shift, Group contrastive loss.

I. INTRODUCTION

Image Quality Assessment (IQA) is crucial for objectively predicting an image's perceptual quality, aligning with the human visual system's perception. It is also used as a loss function in deep learning models for various image processing and computer vision tasks. Methods such as those proposed by [1]–[3] use deep features or semantic cues to assess image quality, while others rely on low-level features like local distortions [4], [5]. Despite these advances, the performance of most IQA models is limited when evaluating images which have a distribution different from the training dataset.

Test-Time Adaptation (TTA) has emerged as an effective approach to handle distribution shifts between training and testing datasets. TTA methods adjust a limited set of model parameters in an unsupervised manner during the testing phase to adapt the baseline model to specific datasets. In literature a few methods are proposed for TTA such as augmentation-based TTA [6], pseudo-labeling [7], [8], and entropy minimization [9] have demonstrated significant potential in enhancing model performance across various tasks. Since labels for testing data are unavailable during TTA, auxiliary tasks are employed to facilitate adaptation. These include aligning feature similarities within a group [10] or across previous test samples [7], generating pseudo-class labels [11], creating augmented views [12], [13], and applying data augmentations such as color transformations [14]. These strategies help compute the pseudo-contrastive loss to adapt the batch normalization layers. At the same time, TTA methods explicitly designed for Image Quality Assessment (IQA) tasks are very limited, and the TTA strategies mentioned above cannot be directly applied to IQA. Recently, Roy et al. [10] introduced a TTA method for no-reference IQA that utilizes pseudo auxiliary tasks such as group contrastive (GC) and rank loss to adjust batch normalization layers. To estimate the GC loss, the feature distance between clusters of low-quality and high-quality images is calculated. These clusters are formed based on the quality score predictions by the base model. This IQA-TTA method faces challenges when the training and testing data distribution differs significantly. In such cases, the base model may struggle to generate accurate image clusters, leading to erroneous GC loss estimation and limited performance. With this in mind, there is a need for a more effective clustering algorithm for IQA-TTA and a new IQA-TTA algorithm.

This paper introduces a novel TTA approach for no-reference IQA that utilizes pre-trained deep learning models, specifically VGG-16, to improve perceptual feature grouping. The proposed method incorporates two auxiliary loss functions—feature affinity-based group contrastive loss and adaptive rank loss—to enable adaptation during inference. The



Fig. 1: Clustering a batch of images from the LIVE dataset using the source model (MetaIQA) highlights misclassified samples in red, resulting in suboptimal performance of the GC loss.

major contributions of the proposed work are as follows:

- 1) We propose a feature affinity-based group contrastive loss framework that leverages pre-trained features and attention mechanisms, ensuring robustness to dataset distribution shifts.
- 2) We propose an adaptive rank-based metric that reduces the impact of rank loss during model adaptation when the base model can effectively distinguish an image from its distorted versions.
- 3) The proposed TTA algorithm demonstrates improved performance compared to existing methods, showing consistent enhancements over state-of-the-art TTA approaches for IQA across various datasets.

The article is structured as follows: Section II covers the motivation, Section III details the proposed losses, Section IV presents experiments, and Section V concludes with future directions.

II. MOTIVATION

The IQA-TTA approach [10] utilizes group contrastive and rank losses to adapt the parameters of batch normalization layers, relying on the base model's ability to classify images into high- and low-quality clusters. However, this method assumes that the base model can accurately predict the relative quality scores for unseen data and subsequently create the clusters of low- and high-quality images to compute the GC loss. This assumption fails when there is a significant distribution shift between the training and testing datasets.

For example, Fig. 1 shows 8 images from the LIVE dataset, along with their corresponding mean opinion scores (MOS) and predicted quality scores using the MetaIQA algorithm [15]. The MetaIQA algorithm is trained on the TID2013 and KADID-10K datasets. From this figure, it is evident that the high-quality image 'Hat' is misclassified as low-quality, while the low-quality image 'Butterfly' is misclassified as high-quality. This results in inaccurate cluster formation, leading to incorrect contrastive loss calculation and reduced

performance. This underscores the need for a more effective clustering method to ensure accurate contrastive loss computation and improve the generalization of IQA-TTA across different datasets.

III. THE PROPOSED ALGORITHM

This section presents a novel TTA algorithm for no-reference IQA. The proposed method incorporates two auxiliary tasks: feature affinity-based perceptual group contrastive loss and adaptive rank loss, to adapt the feature extractor of a pre-trained IQA model during inference. These tasks aim to align the training and testing data distributions without requiring labeled data, enabling effective quality assessment even under distributional shifts. The details of the proposed algorithm are given in the following subsections:

A. Problem statement

Following the framework of IQA-TTA [10], we consider a source/base model, denoted as S_θ comprising a feature extractor (θ_f) and a quality regressor (θ_r). TTA focuses on updating only the feature extractor θ_f during inference, aligning it with the test data distribution while preserving the characteristics of the training distribution. To facilitate this update, a projection head θ_P is added with the feature extractor (θ_f) to project extracted features into a lower-dimensional space. The following loss function is utilized to jointly optimize θ_f and θ_P , leveraging auxiliary losses in the absence of true labels.

$$\theta_f^*, \theta_P^* = \arg \min_{\theta} \mathcal{L}(\theta)$$

The loss function, $\mathcal{L}$ is formulated as a combination of the proposed feature affinity-based group contrastive loss and adaptive rank loss, where $\theta = \{\theta_f, \theta_P\}$ represents the parameters being optimized for the test data.

B. Feature Affinity-based Group Contrastive (FAGC) loss

The overall pipeline of the Feature Affinity-based Group Contrastive (FAGC) loss is illustrated in Fig. 2. This proposed loss enhances the separability between high- and low-quality image clusters. In the IQA-TTA method, a mini-batch of images is divided into two clusters: C_1 , representing low-quality images, and C_2 , representing high-quality images, based on their predicted scores from the base model. The group contrastive (GC) loss is then calculated by maximizing the feature distance between these two clusters.

Notably, the IQA-TTA [10] model forms clusters based on predictions made by the base model, which may not perform efficiently on the testing data, resulting in improperly formed clusters. At the same time, it can be assumed that images with richer high-level information possess better perceptual quality. Motivated by this observation, we propose a new clustering method for GC loss calculation leveraging the pre-trained VGG-16 architecture. For a given set of images, $\{x_1, x_2, \dots, x_N\}$, is fed to the base model S to calculate the quality scores, and here, N is the batch size. Let i_{max} be the index of the image with the best perceptual quality

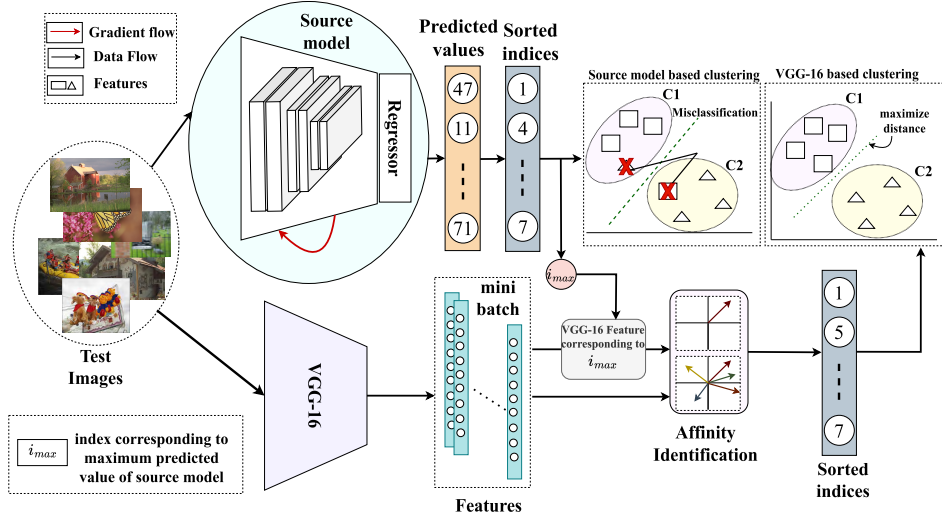


Fig. 2: The proposed feature affinity-based group contrastive loss uses the highest quality image from the source model, which is passed through VGG-16 to extract features. The features of the other samples in the mini-batch are projected into feature space to compute affinity scores, which are then used to sort and cluster the samples.

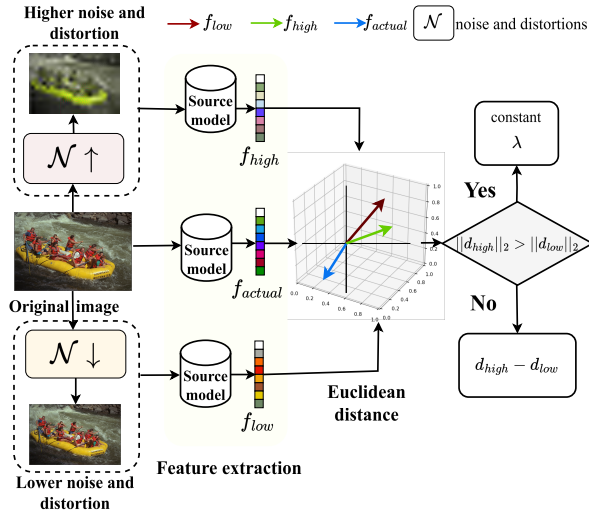


Fig. 3: Proposed adaptive rank loss used for test time adaptation.

estimated using the base model S . The same set of images $\{x_1, x_2, \dots, x_N\}$ are fed to pre-trained VGG-16 to extract the perceptual features $\{f_1, f_2, \dots, f_N\}$. Similarly, $f_{i_{max}}$ represents the VGG-16 features of the image with the highest perceptual quality. If another image exhibits similar feature affinity with $f_{i_{max}}$, it can be inferred that the image also contains rich high-level information and can be regarded as high quality. The feature affinity (aff) between $f_{i_{max}}$ and remaining of the images in a batch is calculated using cosine similarity as:

$$aff(f_{i_{max}}, f_i) = \frac{f_{i_{max}} \cdot f_i}{\|f_{i_{max}}\| \|f_i\|}, \quad \forall i \in N, i \neq i_{max} \quad (1)$$

Images with higher feature affinity with the highest perceptual quality image are classified into high-quality clusters (C_1), while those with lower feature affinity are grouped into low-quality clusters (C_2). These clusters are employed to calculate the Feature Affinity-based Group Contrastive (FAGC) loss by maximizing the feature distance between images in opposite clusters and minimizing the feature distance between images within the same cluster. The FAGC loss for the i^{th} sample in a high-quality cluster (C_1), with respect to the j^{th} sample in same cluster and other k^{th} samples in a low-quality image cluster (C_2), is computed as:

$$\mathcal{L}_{FAGC} = -\log \left(\frac{\exp \left(\frac{Sim(l_i, l_j)}{\tau} \right)}{\sum_{k=(N/2)+1}^N \exp \left(\frac{Sim(l_i, l_k)}{\tau} \right)} \right) \quad (2)$$

Here, l_i and l_j represent the features extracted from the projection head (θ_P) for the i^{th} and j^{th} samples, and $Sim(l_i, l_j)$ denotes the similarity between these features, calculated using the attention between them. Additionally, τ is a temperature parameter that controls the sharpness of the distribution, thereby enhancing the model's ability to effectively separate the clusters. Similarly, the $\mathcal{L}_{FAGC}$ loss is also computed for the other samples in clusters C_1 and C_2 .

C. Adaptive Rank loss

The GC loss struggles to perform effectively when images in a batch exhibit similar perceptual quality [10], as efficient clusters may not be formed. On the other hand, the rank loss assumes that the features extracted by an efficient quality assessment algorithm from the feature extraction module (θ_f) of an image will be less affected by lower noise and more significantly altered by higher noise. If the base model violates this assumption, it indicates that the base model cannot

TABLE I: Summary of datasets used for TTA of IQA metrics.

Dataset	Description
LIVE [16]	29 reference images distorted with various factors, generating 779 distorted images.
CID 2013 [17]	Images captured from 79 imaging devices, totaling 480 images.
KONIQ-10K [18]	Approximately 10k images captured in the wild, featuring natural distortions.
SPAQ [19]	11,125 images captured by smartphones, also featuring natural distortions.

distinguish between slightly noisy and heavily noisy images, thus requiring adaptation. Conversely, when the base model can effectively differentiate between less noisy and noisy images, adaptation is no longer needed. However, the IQA-TTA [10] continues to update the base model (upto some extent) regardless of the base model's ability to differentiate between images with varying perceptual qualities. In light of this, we propose a new adaptive rank mechanism that applies the rank loss selectively, depending on the base model's ability to distinguish between images with varying perceptual quality.

Let f_{high} , f_{low} , and f_{actual} represent features (extracted from the feature extraction module θ_P of the base model) of highly degraded, mildly degraded, and original images, respectively. In an ideal situation when base model is efficient enough, the following relationship between these features should exist:

$$\underbrace{\|f_{high} - f_{actual}\|_2}_{d_{high}} > \underbrace{\|f_{low} - f_{actual}\|_2}_{d_{low}} \quad (3)$$

The proposed adaptive rank loss is based on the observation that if, in most scenarios, the base model can effectively distinguish between highly and less distorted images, then adaptation using the rank loss is not required, and vice versa. With this in mind, we propose counting (using counters *counter1* and *counter2*) the number of times the rank loss, as an auxiliary loss, is required ($|d_{high}| - |d_{low}| > 0$) or not required ($|d_{high}| - |d_{low}| \leq 0$). The adaptive term (γ) for the rank loss is defined as:

TABLE II: Summary of Models addapted Using the proposed TTA Method.

Model	Training Dataset
MetaIQA [15]	Trained on TID2013 [20] and KADID-10K dataset [21].
HyperIQA [22]	Trained on LIVEFB database [23].
TReS [24]	Trained on LIVEFB database [23].
MUSIQ [25]	Trained on LIVEFB database [23].

$$\gamma = \begin{cases} \lambda, & \text{if } counter1 \geq counter2 \\ d_{high} - d_{low}, & \text{otherwise} \end{cases} \quad (4)$$

Where λ is a constant term used to alleviate the impact of the rank loss during TTA. We empirically set λ to one, and changing this value does not significantly affect the overall performance of the proposed algorithm. The adaptive rank loss for a batch of size N is defined as:

$$\mathcal{L}_{Adaptive_Rank} = - \sum_{i=1}^N \log(y_i) \quad (5)$$

Here y is the predicted probability which is calculated as:

$$y = P(d_{high} \geq d_{low}) = \frac{\exp(\gamma)}{1 + \exp(\gamma)} \quad (6)$$

Here, it is evident from equation (4) that the inclusion of the parameter γ allows the model to adaptively determine when the rank loss is required during the adaptation of base model.

D. Total loss

The final loss required to adapt the based model is a weighted combination of the proposed FAGC ($\mathcal{L}_{FAGC}$) and the adaptive rank ($\mathcal{L}_{Adaptive_Rank}$) losses:

$$\mathcal{L} = \alpha_1 \cdot \mathcal{L}_{FAGC} + \alpha_2 \cdot \mathcal{L}_{Adaptive_Rank} \quad (7)$$

Where α_1 and α_2 are weights that determine the contributions of the FAGC and Adaptive rank losses, respectively. As suggested by IQA-TTA [10], these parameters are empirically set to one.

IV. EXPERIMENTS AND RESULTS

This section presents the implementation details of the proposed method, followed by a comprehensive performance evaluation and a comparison with state-of-the-art methods. Furthermore, an ablation study is conducted to analyze the individual contributions of the proposed FAGC and adaptive rank losses. The proposed algorithm is also exclusively compared with IQA-TTA [10], as it is, to the best of our knowledge, the most recent method in the literature that applies Test-Time Adaptation (TTA) for image quality assessment metrics.

A. Implementation details

The proposed model is implemented on an NVIDIA A100 GPU with 80 GB of RAM, utilizing the PyTorch framework on a Linux-based operating system. A batch size of 8 is used to update the batch normalization layers, while the rest of the base model parameters/hyperparameters remain unaltered. The proposed algorithm is applied to perform TTA for four IQA algorithms (details provided in Table II) and is evaluated on four datasets (details provided in Table I) across these IQA algorithms. For MetaIQA, the base model without adaptation processes a batch of eight images in 5.603 ms, IQA-TTA [10] takes approximately 6.846 ms, and our proposed method takes around 7.204 ms.

B. Performance Analysis

To assess the performance of the proposed method, we utilize correlation metrics such as Spearman's Rank Order Correlation Coefficient (SRCC) and Pearson's Linear Correlation Coefficient (PLCC). SRCC measures the consistency of ranking between samples, while PLCC evaluates the agreement between predicted scores and the Mean Opinion Scores (MOS) in the dataset. Table III provides a quantitative comparison of the results for Group Contrastive [10], the proposed

TABLE III: Performance comparison of the proposed TTA algorithm with existing TTA methods across multiple algorithms and datasets. Here $\uparrow$ indicates that a higher value is better, while **red** and **blue** denote the best and second-best performances. Additionally, GC and FAGC are abbreviations for Group Contrastive and Feature Affinity-Based Group Contrastive Loss.

Algorithm $\downarrow$	Dataset $\rightarrow$	Live Dataset [16]		SPAQ Dataset [19]		CID [17]		KONIQ [18]		Average	
		SRCC $\uparrow$	PLCC $\uparrow$	SRCC $\uparrow$	PLCC $\uparrow$	SRCC $\uparrow$	PLCC $\uparrow$	SRCC $\uparrow$	PLCC $\uparrow$	SRCC $\uparrow$	PLCC $\uparrow$
MetaIQA [15]	Base	0.73231	0.67319	0.59636	0.54684	0.72126	0.68168	0.52097	0.47602	0.64273	0.60409
	GC only [10]	0.75668	0.69291	0.62566	0.57872	0.78631	0.70678	0.54506	0.50435	0.67843	0.63224
	Rank only [10]	0.76708	0.73120	0.63955	0.60707	0.75564	0.72246	0.53441	0.51042	0.67417	0.64906
	GC+ Rank [10]	0.78986	0.76469	0.60067	0.57503	0.77227	0.73644	0.55492	0.53278	0.67943	0.65767
	FAGC (proposed)	0.81800	0.75719	0.68807	0.64260	0.76931	0.70916	0.55496	0.51426	0.70758	0.66616
	Adaptive rank (proposed)	0.79887	0.76708	0.65004	0.61722	0.75638	0.71399	0.55496	0.52895	0.69006	0.66346
	Overall (Proposed)	0.85206	0.83264	0.70545	0.69027	0.77201	0.73212	0.57355	0.55288	0.72577	0.70674
HyperIQA [22]	Base	0.51316	0.44358	0.59577	0.56162	0.53247	0.65048	0.59406	0.63612	0.55887	0.57013
	GC only [10]	0.50576	0.43590	0.69863	0.64298	0.48519	0.59058	0.64699	0.68415	0.58414	0.58755
	Rank only [10]	0.56668	0.52819	0.62146	0.59352	0.50852	0.61670	0.61441	0.66542	0.57777	0.59632
	GC+ Rank [10]	0.53721	0.49698	0.60333	0.57319	0.61595	0.67336	0.61963	0.66967	0.59403	0.60144
	FAGC (proposed)	0.61203	0.53866	0.74663	0.66542	0.52178	0.61812	0.56363	0.61542	0.61102	0.60973
	Adaptive rank (proposed)	0.56442	0.51907	0.68022	0.62055	0.50863	0.61127	0.62301	0.66902	0.59407	0.60280
	Overall (Proposed)	0.58841	0.55186	0.61012	0.58167	0.56660	0.65946	0.61856	0.66699	0.59592	0.61118
TReS [24]	Base	0.54340	0.44499	0.70137	0.58539	0.53247	0.65048	0.66896	0.71042	0.61155	0.60056
	GC only [10]	0.69484	0.61707	0.70883	0.65216	0.56397	0.67087	0.64109	0.69844	0.63330	0.65492
	Rank only [10]	0.63626	0.58653	0.71887	0.62782	0.58973	0.66756	0.67117	0.71985	0.65401	0.65115
	GC+ Rank [10]	0.66124	0.62124	0.73341	0.65357	0.61595	0.67336	0.64682	0.70063	0.66435	0.66263
	FAGC (proposed)	0.76628	0.70189	0.77507	0.65696	0.54450	0.66179	0.62221	0.68232	0.67702	0.67599
	Adaptive rank (proposed)	0.61155	0.56062	0.74663	0.65410	0.58773	0.67059	0.69244	0.73847	0.65959	0.65667
	Overall (Proposed)	0.75962	0.66970	0.75364	0.67820	0.56660	0.65946	0.69174	0.73811	0.69290	0.68768
MUSIQ [25]	Base	0.27867	0.35615	0.78694	0.74800	0.53207	0.60643	0.64586	0.66822	0.56088	0.58794
	GC only [10]	0.33566	0.39592	0.78442	0.73603	0.54784	0.61864	0.67063	0.72507	0.58464	0.61206
	Rank only [10]	0.34355	0.39782	0.78083	0.73045	0.56233	0.62934	0.66638	0.72215	0.58827	0.61361
	GC+ Rank [10]	0.33452	0.39478	0.78224	0.73232	0.56264	0.62658	0.66822	0.72193	0.58691	0.61250
	FAGC (proposed)	0.37017	0.41219	0.78275	0.73340	0.55018	0.62185	0.55488	0.51420	0.56491	0.56984
	Adaptive rank (proposed)	0.35930	0.40757	0.78204	0.73200	0.54753	0.61803	0.67001	0.72522	0.58972	0.61451
	Overall (Proposed)	0.35609	0.40627	0.78417	0.73435	0.54689	0.61757	0.67113	0.72627	0.58957	0.61480

FAGC, Rank [10], the proposed Adaptive Rank, IQA-TTA [10], and the proposed overall algorithm. The results in Table III demonstrate that the proposed TTA method achieves better performance overall compared to the IQA-TTA [10] approach. These findings demonstrate the effectiveness and efficiency of the proposed algorithm.

C. Ablation study

An ablation study in Tables IV and V assesses the impact of feature extraction methods (such as pre-trained deep-learning methods) and similarity metrics. Among feature extraction methods, pretrained VGG-16 outperformed others on the LIVE [16], CID [17], and KONIQ-10K [18] datasets in terms of PLCC and SRCC. Cosine similarity in the latent space proved to be the most effective feature affinity metric, as shown in Table V. Additionally, the scatter plots in Fig. 4 demonstrate that the predicted scores from the proposed algorithm align more closely with the regression line, highlighting its effectiveness.

V. CONCLUSION

In this work, we proposed a new TTA algorithm for IQA tasks, incorporating two auxiliary tasks: feature affinity-based group contrastive loss (FAGC) and adaptive rank loss. The FAGC loss, based on pre-trained deep learning models, efficiently clusters low and high-quality images. Additionally, adaptive rank loss mitigates the impact of rank loss when the base model distinguishes images with varying distortion levels. The proposed algorithm is tested with multiple algorithms

across several datasets, outperforming existing methods in TTA tasks. One limitation of the proposed work is that it does not perform satisfactorily when the base model is unable to accurately identify images with the best perceptual quality. An

TABLE IV: The ablation study to analyze the impact of different feature extraction algorithms on the proposed Feature Affinity based Group Contrastive (FAGC) loss. **Red** indicates the best performance. Here, FEM stands for the Feature Extraction Method.

Model	FEM	Live [16]		CID [17]		KONIQ [18]		Average	
		SRCC	PLCC	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC
MetaIQA [15]	Base	0.732	0.673	0.721	0.681	0.520	0.476	0.658	0.610
	VGG-16	0.852	0.832	0.772	0.732	0.573	0.552	0.732	0.705
	VGG-19	0.688	0.684	0.611	0.538	0.447	0.447	0.582	0.557
	ResNet-18	0.801	0.779	0.747	0.711	0.594	0.566	0.714	0.685
	ResNet-50	0.850	0.831	0.781	0.740	0.560	0.535	0.643	0.702
	ViT	0.833	0.811	0.765	0.728	0.559	0.537	0.719	0.692
HyperIQA [22]	InceptionV3	0.716	0.689	0.701	0.646	0.414	0.398	0.611	0.578
	Base	0.513	0.443	0.503	0.614	0.594	0.636	0.536	0.564
	VGG-16	0.588	0.551	0.535	0.608	0.618	0.666	0.580	0.609
	VGG-19	0.587	0.551	0.531	0.611	0.623	0.670	0.580	0.611
	ResNet-18	0.610	0.569	0.519	0.612	0.666	0.621	0.598	0.601
	ResNet-50	0.608	0.565	0.518	0.612	0.619	0.667	0.582	0.615
TReS [24]	ViT	0.590	0.552	0.516	0.605	0.618	0.667	0.575	0.608
	InceptionV3	0.589	0.551	0.496	0.592	0.574	0.678	0.553	0.607
	Base	0.543	0.444	0.532	0.650	0.668	0.710	0.581	0.601
	VGG-16	0.759	0.669	0.566	0.659	0.691	0.738	0.672	0.689
	VGG-19	0.631	0.580	0.587	0.661	0.609	0.663	0.609	0.635
	ResNet-18	0.675	0.615	0.597	0.678	0.647	0.712	0.639	0.668
TReS [24]	ResNet-50	0.673	0.614	0.599	0.679	0.655	0.716	0.636	0.647
	ViT	0.673	0.615	0.596	0.678	0.649	0.712	0.639	0.668
	InceptionV3	0.592	0.540	0.580	0.660	0.699	0.744	0.586	0.600

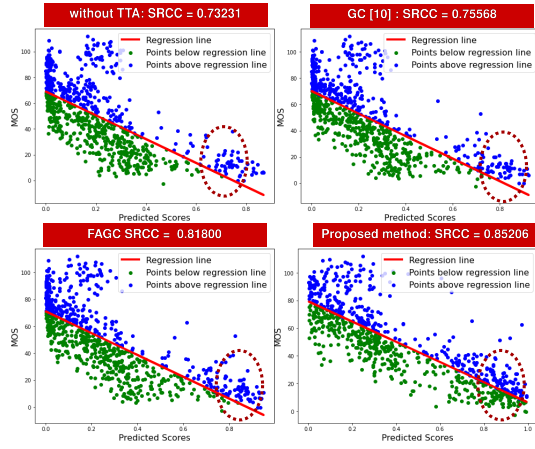


Fig. 4: Scatter plot showing the relationship between the MOS and predicted scores using the base model, with GC loss, proposed FAGC, and the overall proposed TTA algorithm on the Live dataset [16], with MetaQA [15] as the source model.

TABLE V: Ablation study examines the impact of different feature affinity calculation methods on the proposed Feature Affinity-based Group Contrastive (FAGC) loss. Red denotes the best performance. CS: Cosine Similarity, ED: Euclidean Distance, PLCC: Pearson’s Linear Correlation Coefficient.

Model	Method	Live [16]		CID [17]		KONIQ [18]		Average	
		SRCC	PLCC	SRCC	PLCC	SRCC	PLCC	SRCC	PLCC
MetaQA [15]	CS	0.852	0.832	0.772	0.732	0.573	0.552	0.732	0.705
	ED	0.827	0.811	0.733	0.737	0.585	0.621	0.715	0.723
	PLCC	0.735	0.702	0.570	0.526	0.406	0.381	0.570	0.536
HyperQA [22]	CS	0.588	0.551	0.535	0.608	0.618	0.666	0.580	0.609
	ED	0.588	0.552	0.517	0.611	0.618	0.666	0.574	0.610
	PLCC	0.587	0.550	0.543	0.617	0.616	0.665	0.582	0.611
TReS [24]	CS	0.759	0.669	0.566	0.659	0.691	0.738	0.672	0.689
	ED	0.672	0.613	0.595	0.677	0.657	0.719	0.641	0.670
	PLCC	0.674	0.615	0.599	0.680	0.651	0.715	0.642	0.670

ensembling or ranking-based approach could be incorporated to mitigate these limitations.

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